

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5

8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

INTRODUCTION TO DBMS:

- DBMS stands for **Database Management System**.
- It is software used to create, store, manage, and retrieve data.
- It provides an easy way to organize and access information.
- It helps in managing both small and large volumes of data.
- It ensures that data is accurate, consistent, and reliable.
- It reduces data redundancy (duplicate data) and maintains data integrity.

OBJECTIVES

- Store data efficiently.
- Reduce data redundancy (duplicate data).
- Ensure data security and privacy.
- Maintain data integrity and accuracy.
- Support multiple users at the same time.
- Enable quick data retrieval and updates.

REAL - WORLD APPLICATIONS

- **Banking** – Customer accounts and transactions.
- **Hospitals** – Patient records and appointments.
- **Schools/Colleges** – Student and faculty information.
- **E-commerce** – Product, order, and customer management.
- **Airline Reservation** – Flight booking and ticket management.
- **Library Management** – Book records and issue/return details.

FILE - BASED SYSTEM VS DBMS

- **File-Based System:**
 - Data stored in separate files.
 - High data redundancy (duplicate records).
 - Difficult to update customer information.
 - Slow data retrieval.
 - Limited security.
- **DBMS:**
 - Data stored in a centralized database.
 - Reduces data redundancy.
 - Easy and fast data updates.
 - Quick data retrieval using queries.
 - Strong security and access control.

ADVANTAGES

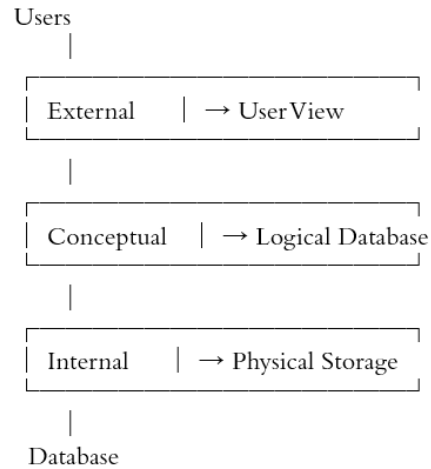
- Minimizes data duplication.
- Improves data security.
- Provides data backup and recovery.
- Allows easy data sharing.
- Increases data consistency.
- Reduces application development time.

FILE - BASED SYSTEM VS DBMS

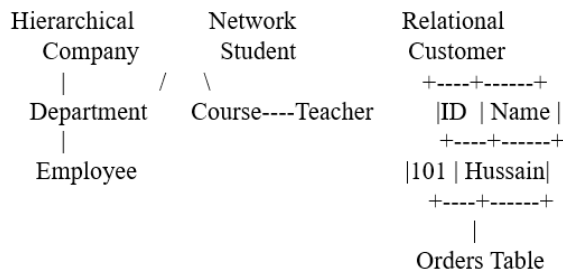
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ANSI/SPARC THREE-LEVEL ARCHITECTURE

- **Levels:**
- **External Level:** Different views for different users.
- **Conceptual Level:** Overall logical structure of the database.
- **Internal Level:** How data is physically stored.
- **Example**
- **Banking System:**
- **Customer:** Views account balance.
- **Manager:** Views all customer records.
- **Database Server:** Stores data on disk.
- **Benefits**
- Data independence
- Better security



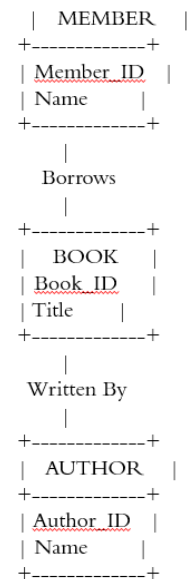
DATA MODELS (HIERARCHICAL, NETWORK, RELATIONAL)



<i>Data Model</i>	<i>Structure</i>	<i>Use Case</i>
<i>Hierarchical</i>	Tree	Banking
<i>Network</i>	Graph	Airline Reservation
<i>Relational</i>	Tables	E-commerce / Hospital

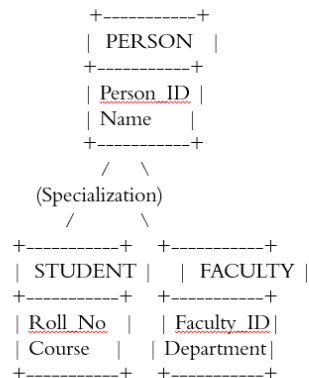
ER DIAGRAM – ONLINE LIBRARY SYSTEM

- **Components**
- **Entities:** Member, Book, Author
- **Primary Keys (PK):** Member_ID, Book_ID, Author_ID
- **Attributes:** Name, Email, Title, ISBN, Author Name
- **Relationship 1:** Member **Borrows** Book (1 Member → Many Books)
- **Relationship 2:** Author **Writes** Book (1 Author → Many Books)
- **Purpose:** Stores and manages library members, books, borrowing records, and author details.



EER MODEL – GENERALIZATION & SPECIALIZATION (UNIVERSITY CASE STUDY)

- **EER (Enhanced Entity-Relationship) Model:** Extends the ER model with advanced features.
- **Generalization:** Combines **Student** and **Faculty** into a common entity **Person**.
- **Specialization:** Splits **Person** into **Student** and **Faculty** based on roles.
- **Common Attributes:** Person_ID, Name.
- **Student Attributes:** Roll_No, Course.
- **Faculty Attributes:** Faculty_ID, Department.



DATA INDEPENDENCE IN DBMS

- **Definition:**
- Data independence means changing the database schema **without affecting application programs**.
- **Logical Data Independence:**
- Changes in the logical structure.
- Does not affect user applications.
- Example: Adding a new column to a table.
- **Physical Data Independence:**
- Changes in physical storage.
- Does not affect the logical schema.
- Example: Changing file organization or adding indexes.
- **Significance:**
- Reduces application maintenance.
- Improves flexibility.
- Increases data security.

SOFTWARE COMPONENTS OF A DBMS

- **Query Processor:**
- Interprets SQL queries.
- Optimizes query execution.
- Retrieves the required data efficiently.
- **Storage Manager:**
- Manages data stored on disk.
- Handles file organization and indexing.
- Controls data access and storage.
- **Transaction Manager:**
- Ensures reliable transaction execution.
- Maintains data consistency and integrity.

STRONG VS WEAK ENTITIES IN ER MODEL

- **Strong Entity:**
 - Has its own Primary Key.
 - Exists independently.
 - Does not depend on another entity.
 - Example: Student, Employee, Customer.
- **Weak Entity:**
 - Does not have a complete Primary Key.
 - Depends on a strong entity.
 - Identified using the strong entity's key.
 - Example: Dependent, Order Item, Student's Family Member.
- Real-World Example
 - Strong Entity: Student (Student_ID, Name)
 - Weak Entity: Dependent (Dependent_Name, Student_ID)

SMALL EER DIAGRAM – RETAIL CHAIN MANAGEMENT SYSTEM

- Supertype: Person
- Subtypes: Customer, Employee
- Store manages customers and employees.
- Store sells products.
- Demonstrates specialization (Person → Customer, Employee).

